Northeast-trending Early Proterozoic dykes of southern Superior Province: multiple episodes of emplacement recognized from integrated paleomagnetism and U–Pb geochronology[1]

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Integrated paleomagnetic and U–Pb geochronologic studies have been conducted to establish the paleomagnetic directions and ages of Early Proterozoic tholeiitic dykes of northeast trend in the southern Superior Province, previously referred to collectively as Preissac dykes. It is demonstrated that they are readily separated on the basis of paleomagnetism into subsets, referred to as the Biscotasing and Senneterre swarms. In addition a pair of unnamed dykes may be associated with the north- and northwest-trending Matachewan swarm farther west.

Biscotasing dykes have a down-west magnetization of single polarity with a corresponding paleopole at 27.8°N, 136.7°W (dm = 12.3° and dp = 9.4°). Senneterre dykes carry an up-north (or occasionally down-south) direction with corresponding paleopole at 15.3°S, 75.7°W (dm = 7.0°, dp = 4.4°). The Senneterre direction is indistinguishable from the primary N1 remanence direction that dominates the magnetization of Nipissing sills of the Southern Province. Paleomagnetic field tests described herein or in earlier studies indicate that Biscotasing and Senneterre directions are primary and, hence, that two ages of intrusion are involved, with the age of Senneterre dykes coinciding with the intrusion of most Nipissing sills. U–Pb dating of baddeleyite conducted at a paleomagnetic sampling site yields an age of 2214.3 ± 12.4 Ma for the Senneterre swarm, indistinguishable from the age of 2217.2 ± 4 Ma reported from an N1 Nipissing sill site in another study. A U–Pb age on baddeleyite and zircon of 2166.7 ± 1.4 Ma was obtained from a paleomagnetic site in the Biscotasing swarm. The primary paleopoles for the Senneterre, Nipissing, and Biscotasing rocks define a direction of polar wander opposite to that of the most widely used polar wander paths for North America for this period, suggesting that these paths should no longer be used.

Des études de paléomagnétisme et de géochronologie avec la méthode U–Pb ont été réalisées conjointement pour déterminer les directions paléomagnétiques et les âges des dykes tholéiitiques orientés vers le nord-est, du Protérozoïque précoce, dans la province du Supérieur méridionale, lesquels étaient autrefois désignés collectivement comme les dykes de Preissac. L'étude paléomagnétique démontre qu'ils sont aisément séparables en deux sous-groupes, appelés ici les essaims de Biscotasing et de Senneterre. En outre une paire de dykes non nommée est possiblement associée à l'essaim de Matachewan de direction nord et nord-ouest apparaissant plus à l'ouest.

Les dykes de Biscotasing fournissent une aimantation de polarité simple de direction ouest vers le bas, avec un paléopôle correspondant à 27,8°N, 136,7°W (dm = 12,3°, dp = 9,4°). Les dykes de Senneterre portent une direction de direction nord vers le haut (ou occasionellement de direction sud vers le bas) avec un paléopôle correspondant à 15,3°S, 75,7°W (dm = 7,0°, dp = 4,4°). Il est impossible de distinguer la direction des dykes de Senneterre de celle de la rémanence primaire N1 qui prédomine dans l'aimantation des filons-couches de Nipissing de la province du Sud. Les tests de champ paléomagnétique décrits ici, ou dans les études antérieures, indiquent que les directions pour les sous-groupes de Biscotasing et de Senneterre sont primaires et, que par conséquent, il y a deux âges d'intrusion, avec un âge pour les dykes de Senneterre qui coïncide avec l'intrusion de la plupart des filons-couches de Nipissing. La datation U–Pb sur baddeleyite prélevée à un site d'échantillonnage paléomagnétique a fourni pour l'essaim de Senneterre un âge de 2214,3 ± 12,4 Ma, indiscernable de l'âge de 2217,2 ± 4 Ma rapporté dans une autre étude pour un site de filon-couche de Nipissing de rémanence N1. Un âge U–Pb sur baddeleyite et zircon de 2166,7 ± 1,4 Ma a été obtenu pour un site paléomagnétique de l'essaim de Biscotasing. Les paléopôles primaires pour les roches de Senneterre, Nipissing et Biscotasing définissent une direction de dérive polaire opposée à celle qui est généralement utilisée dans les trajets de la dérive des pôles de l'Amérique du Nord pour cette période, ce qui suggère que dorénavant ces trajets ne devraient pas être utilisés.

[Traduit par la rédaction]

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Introduction

The several sets of northeast-trending mafic dykes of the Superior Province southeast of the Kapuskasing Structural Zone (Fig. 1) have frequently been confused in the literature. Perhaps the most important distinction that has previously been made (e.g., Card 1981) is between olivine diabase Abitibi dykes of Middle Proterozoic age (1141 ± 2 Ma; U–Pb baddeleyite; Krogh et al. 1987) and quartz tholeiite Preissac dykes of Early Proterozoic age (2150 ± 50 Ma; $^{40}Ar/^{39}Ar$ mafic minerals; Hanes and York 1979). Yet, many studies still fail to properly separate even these dykes of two vastly different ages. The paleomagnetism of Middle Proterozoic Abitibi dykes has recently been reexamined (Ernst and Buchan 1993), but reliable paleopoles have not been published for northeast-trending Early Proterozoic dykes of the southern Superior Province.

The Early Proterozoic dykes of northeast trend vary in thickness from a few metres to 100 m or more and have subvertical to vertical dips. Because there are relatively few widely spaced dykes, the thickest dykes can be traced for tens or hundreds of kilometres (Figs. 1 and 2). They cut obliquely across Archean and Early Proterozoic lithological units and structural trends in the Superior and Southern provinces, respectively, and are displaced by northwest- and north-northwest-trending faults of the Timiskaming and Onaping sets. They do not parallel major structural elements such as the Kapuskasing Structural Zone and rarely, if ever, occupy regional faults. Fahrig (1987) suggested that emplacement of these dykes, together with the Fort Frances and Marathon swarms farther west, may be related to tectonic events in the Early Proterozoic Penokean Orogen to the south.

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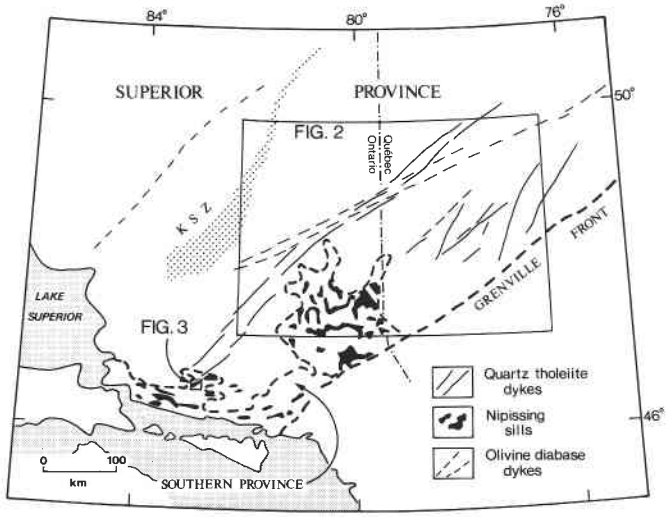


FIG. 1. Sampling localities in the southern Superior and Southern provinces, showing distribution of northeast-trending quartz tholeiite dykes and Nipissing sills of Early Proterozic age and olivine diabase Abitibi dykes of Middle Proterozoic age. KSZ, Kapuskasing Structural Zone.

In this study of paleomagnetism, U−Pb geochronology, and petrography of the Early Proterozoic quartz tholeiite dykes, two (or possibly three) northeast-trending dyke swarms have been identified. They will be referred to as the ~2214 Ma Senneterre swarm (dykes labelled S in Fig. 2) with an up-north (or down-south) remanence, the 2166.7 ± 1.4 Ma Biscotasing swarm (dykes labelled B in Fig. 2) with a down-west remanence, and two unnamed dykes (U1 and U2 of Fig. 2) with a poorly defined shallow south-southwest magnetization that suggests an association with Matachewan dykes emplaced at 2452^{+3}_{-2} Ma (Heaman 1989). For convenience, Table 1 summarizes the characteristics of each of the northeast-trending Early and Middle Proterozoic swarms that occur southeast of the Kapuskasing Structural Zone.

Previous studies

Three distinct directions of paleomagnetic remanence have been reported in the literature. The dominant down-west magnetization was described in a number of studies including Strangway (1961), Fahrig et al. (1965), Larochelle (1966), Irving and Naldrett (1977), and Schutts and Dunlop (1981). However, it is now known (e.g., Buchan et al. 1989, caption to Fig. 12; Ernst and Buchan 1993) that Abitibi dykes of Middle Proterozoic age and Biscotasing dykes of Early Proterozoic age were not differentiated in this early work because both swarms carry down-west remanences.

A second remanence (herein called the Senneterre remanence) has been reported in the literature at only four sites. It was recognized as an up-north magnetization by Larochelle (1966) at three sampling sites in two dykes, and as a magnetization of opposite polarity by Symons et al. (1981) at a single site. Preliminary unpublished work (W.F. Fahrig, personal communication 1985) indicated that this magnetization occurs in other northeast-trending quartz tholeiite dykes.

Finally, a third shallow south-southwest remanence direction was obtained by Larochelle (1966) at four sites from a single dyke. The magnetization of this dyke (declination D = 221°, inclination I = 19°) was used by W.F. Fahrig (in Goodwin et al. 1972, Table 8) to define the characteristic remanence direction of "Preissac" dykes, a term he applied to all Early Proterozoic quartz tholeiite dykes of northeast trend in the southern Superior. As will be seen below, this direction is not typical of most other northeast-trending quartz tholeiite dykes.

Larochelle (1966) interpreted the three remanence directions of northeast-trending dykes to indicate three ages of dyke emplacement. On the other hand, Gates and Hurley (1973), in a Rb−Sr whole-rock dating study that used samples from Larochelle's paleomagnetic sites as well as from three additional sites, concluded that all the dykes, regardless of paleomagnetic direction, are of about the same age (2102 ± 67 Ma using decay constant of 1.42×10^{-11} year^{-1}).

As noted above, more recent geochronologic studies (Krogh et al. 1987; Hanes and York 1979) have clarified the existence of Middle and Early Proterozoic dykes of northeast trend. However, the Early Proterozoic dykes have not been separated into distinct swarms using high-precision U−Pb dating at sites with different paleomagnetic signatures.

Sampling and laboratory procedure

An average of five oriented block samples or drilled cores were collected for paleomagnetic study at each of 35 sites in 14 northeast-trending dykes over a broad geographical area of the southern Superior Province in northeastern Ontario and northwestern Quebec (Fig. 2). Two northeast-trending "Nipissing" dykes that intrude Nipissing sills in the Elliot Lake area of the Southern Province were also sampled (Fig. 3).

Petrographic examination of one or more samples from each site was carried out to demonstrate that each dyke belongs to the Early Proterozoic quartz tholeiite set rather than the olivine diabase Abitibi swarm of Middle Proterozoic age. Baked contact tests were attempted at five localities to determine if the remanence is primary or secondary.

Two or three cylindrical specimens (2.5 cm in diameter and 2.2 cm in length) were cut from each sample and their magnetization measured using a Schonstedt spinner magnetometer. At each site two specimens per sample were alternating-field (AF) demagnetized in about 15 steps to peak fields of 100 mT with a Schonstedt demagnetizer. A few representative specimens were thermally cleaned in about 12 steps to peak temperatures of 580°C using a Schonstedt thermal demagnetizer modified to maintain temperatures within ±5°C.

Stable remanence directions for individual specimens were calculated using the least-squares-fit method of Kirschvink (1980). Specimen directions were then averaged to obtain sample directions, samples directions were averaged to determine site means, and site directions were averaged to obtain means for each dyke.

U−Pb dating of zircon and baddeleyite has proven to be the most reliable method for determining the crystallization age of mafic intrusions (e.g., Corfu and Andrews 1986; Krogh et al. 1987; LeCheminant and Heaman 1989). In the present study, sampling for U−Pb geochronology was conducted at several stably magnetized paleomagnetic sites. We separated baddeleyite and (or) zircon from 25 kg samples using a Rogers gravity table and heavy-liquids techniques. Samples were further concentrated using a Frantz magnetic separator and final hand picking under a binocular microscope. Every attempt was made to exclude baddeleyite grains that were partially altered to secondary zircon; however, because of the very fine grain size of the concentrate (typically <40 μm), this could not always

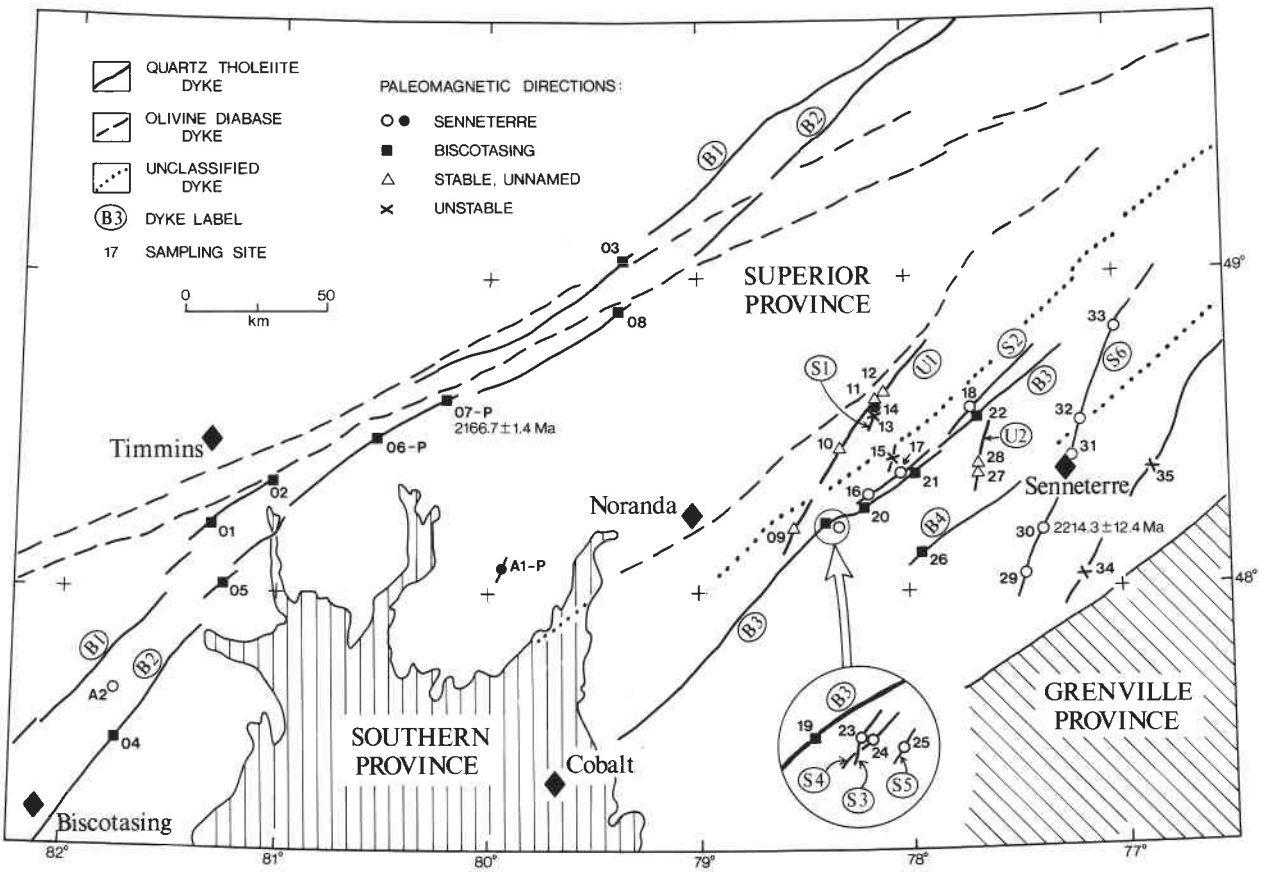


FIG. 2. Paleomagnetic characteristics, positive baked contact tests (labelled P), and U–Pb ages for sampling sites in northeast-trending quartz tholeiite dykes of the southern Superior Province. Biscotasing dykes are labelled B, Senneterre dykes S, and stable, unnamed dykes U. For comparison, olivine diabase dykes of the Abitibi swarm are also shown. Dyke locations and trends are modified after Ministère de l'Énergie et des Ressources du Québec (1981*a*, 1981*b*, 1981*c*) and Ontario Geological Survey (1991). Site A1 indicates the locality of a positive baked contact test for a Senneterre dyke reported by Symons et al. (1981). Site A2 is in a northeast-trending dyke carrying a Senneterre direction (H.C. Halls, personal communication, 1992), which is located at Matachewan dyke site TM34 of Bates and Halls (1991). Normally and reversely magnetized sites are shown as closed and open symbols, respectively.

TABLE 1. Northeast-trending mafic dykes of the southern Superior Province

U–Pb age (Ma)	Average trend	Petrologic type	Paleomagnetic direction	Name in this paper	Former name(s)
1141±2[a]	ENE	Olivine diabase	Down W[b]	Abitibi	Abitibi
2166.7±1.4	NE	Quartz tholeiite	Down W	Biscotasing	Preissac, Abitibi, locally Nipissing dykes
2214.3±12.4	NNE	Quartz tholeiite	Up N	Senneterre	Preissac, Abitibi
?	NNE	Quartz tholeiite	Shallow SSW[c]	?	Preissac, Abitibi

[a]Age of Abitibi swarm is from Krogh et al. (1987).
[b]Paleomagnetic direction of Abitibi swarm is reported in Ernst and Buchan (1993).
[c]Paleomagnetic directions suggest that these dykes may represent a north-northest-trending subswarm of the Matachewan dyke swarm.

be done with complete confidence. Zircon and baddeleyite were analysed at the Geological Survey of Canada laboratory using methods described by Mortensen (1993).

Petrography

For the most part, northeast-trending quartz tholeiite dykes of Early Proterozoic age are readily distinguishable throughout the study area from northeast-trending olivine diabase Abitibi dykes of Middle Proterozoic age (Card 1981; Condie et al. 1987; Ernst and Buchan 1993). The quartz tholeiite dykes are quartz normative, generally contain modal quartz, but lack olivine, plot in the subalkaline field on alkali-silica diagrams, and have little light rare-earth element enrichment. Middle Proterozoic Abitibi dykes, on the other hand, are olivine normative, generally contain modal olivine, and are light rare-earth element enriched.

Quartz tholeiite dykes of the Senneterre and Biscotasing swarms, as well as the pair of unnamed dykes labelled U1 and U2 in Fig. 2, which are separated on the basis of paleomag-

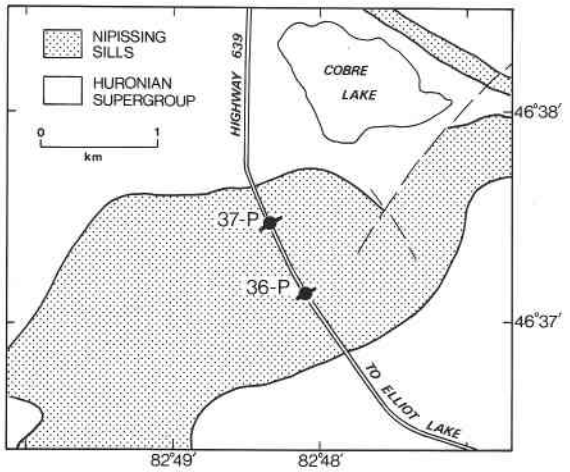


FIG. 3. Sampling sites in the Southern Province north of Elliot Lake. Symbols as in Fig. 2. Sampling site 36-P is located in dyke B5, site 37 in dyke B6. Positive baked contact tests (P) are available at both sites.

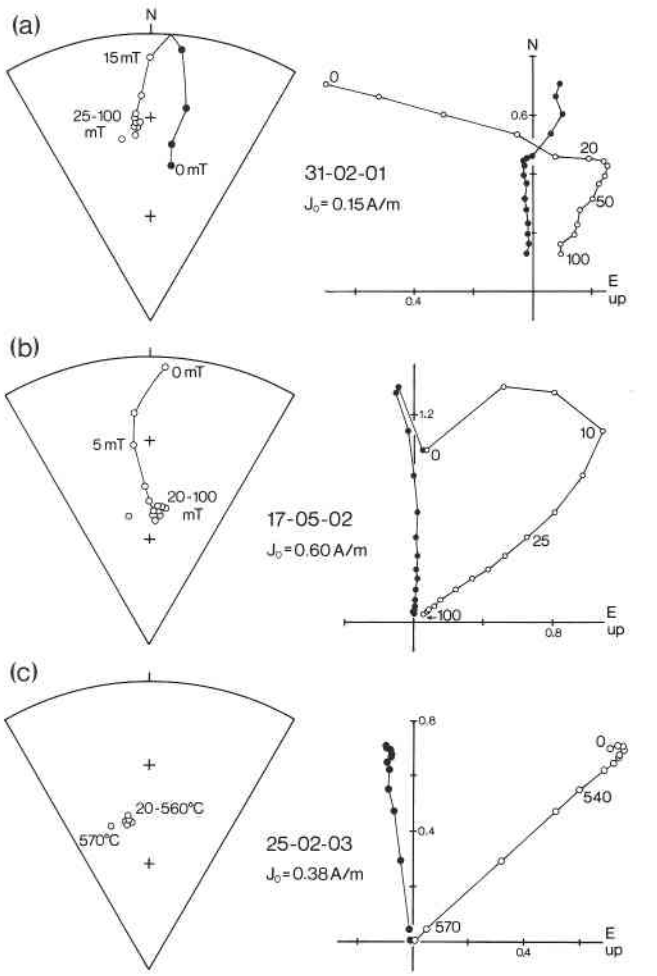


FIG. 4. Paleomagnetic results for Senneterre dyke specimens: (*a*) at site 31 on AF demagnetization, (*b*) at site 17 on AF demagnetization, and (*c*) at site 25 on thermal demagnetization. Directional data are plotted with respect to present horizontal on an equal-area net, with closed and open symbols indicating down and up directions, respectively. Orthogonal-component plots are shown, with closed symbols indicating projection onto the horizontal plane and open symbols indicating projection onto the up and north vertical plane. Orthogonal-component data are plotted relative to initial intensity of natural remanent magnetization, J_0.

netic and geochronologic differences in this paper, could not be distinguished petrographically. All have fine-grained chilled margins and medium- to coarse-grained interiors consisting of approximately 50% plagioclase (labradorite), 30% pyroxene (either clinopyroxene, or clinopyroxene and orthopyroxene), up to 10% quartz, and several percent magnetite–ilmenite intergrowths. Alteration of the dykes includes saussuritization of plagioclase, uralitization and chloritization of pyroxenes, and conversion of the primary oxides to aggregates of leucoxene with remnent magnetite blades. Alteration is highly irregular from one dyke to another, within individual dykes, and, indeed, within individual hand samples. In some samples, pyroxenes are highly altered, whereas primary plagioclases are largely unaffected. In other samples, the converse is true. In yet others, all of the primary minerals are altered. There is no correlation between the degree of alteration and the preservation of primary paleomagnetic remanences, indicating that alteration of the dykes is probably largely deuteric rather than due to later regional metamorphism.

Paleomagnetic results

Stably magnetized dykes fall into three groupings based on paleomagnetic direction (Fig. 2). Senneterre dykes carry an up-north (down-south) remanence. Biscotasing dykes carry a down-west magnetization. Two unnamed dykes (U1 and U2) have a somewhat scattered shallow south-southwest remanence. No attempt was made to classify two other dykes that were unstably magnetized.

There is no evidence of superposition of Senneterre and Biscotasing directions at individual sampling localities. In addition, sampling sites thought to lie along a single dyke always carry a consistent stable direction of magnetization. Examples of paleomagnetic data from Senneterre, Biscotasing, and unnamed dykes U1 and U2 are illustrated in Figs. 4, 5, and 6, respectively.

The up-north (down-south) Senneterre direction (Fig. 4) is isolated at 13 sampling sites in six dykes. Blocking temperatures obtained in thermal cleaning experiments (Fig. 4*c*) indicate that low-titanium titanomagnetite carries the stable magnetization. The Senneterre dykes of this study are confined to the Senneterre–Noranda area (Fig. 2). However, the Senneterre dyke swarm is likely much more extensive. The Senneterre paleomagnetic direction has been observed in a site from a north-northeast-trending dyke about 50 km west of Noranda (Symons et al. 1981) (site A1 of Fig. 2), at a site in a northeast-trending dyke about 100 km south-southwest of Timmins (site A2 of Fig. 2) (H.C. Halls, personal communication, 1992), and at two north-northeast-trending dyke sites in the Lacs Obatogamau area about 230 km northeast of Senneterre (W.F. Fahrig, personal communication, 1985).

The Senneterre direction is indistinguishable from the dominant N1 direction of magnetization reported in a number of studies of Nipissing Diabase sills of the Southern Province (e.g., Symons 1970, 1971). Both the Senneterre and Nipissing N1 remanences are interpreted to be primary on the basis of paleomagnetic field tests as described below.

The down-west Biscotasing magnetization (Fig. 5) is observed at 18 sampling sites along six dykes (Figs. 2 and 3). The two dykes that cut Nipissing sills north of Elliot Lake (Fig. 3) were

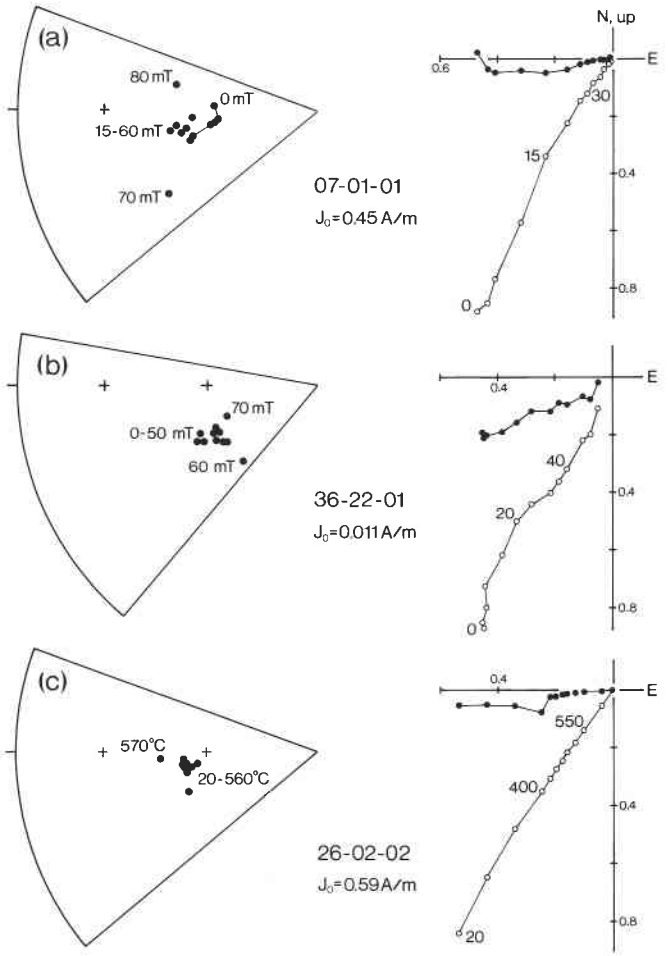


FIG. 5. Paleomagnetic results for Biscotasing dyke specimens: (*a*) at site 7 upon AF demagnetization, (*b*) at site 36 upon AF demagnetization, and (*c*) at site 26 upon thermal demagnetization. Symbols as in Fig. 4. The up-east vertical plane has been used in the orthogonal-component plots.

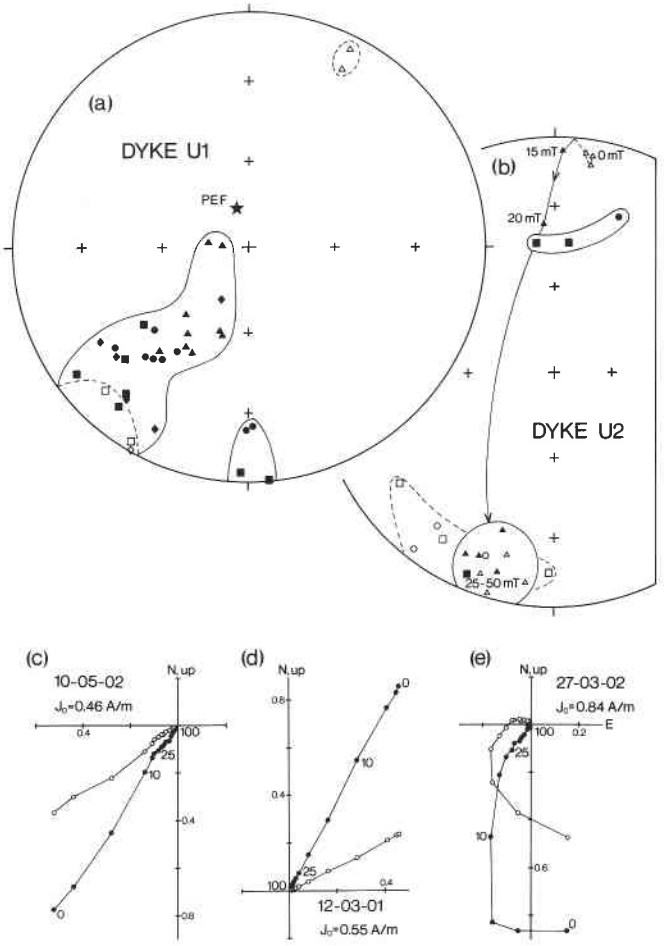


FIG. 6. Paleomagnetic results for unnamed dykes U1 and U2. (*a*) Individual specimen directions, calculated using Kirschvink (1980) analysis, from sites 9 (♦,◇), 10 (■,□), 11 (●), and 12 (▲,△) of dyke U1. The star labelled PEF represents the direction of the present Earth's field. (*b*) Specimen directions from sites 27 (●,○) and 28 (■,□) of dyke U2. ▲,△, directions upon stepwise AF demagnetization of a single specimen from site 28. (*c*), (*d*), and (*e*) Orthogonal-component plots (showing the up-east vertical plane) for specimens from sites 10, 12, and 27, respectively. Symbols as in Fig. 4.

previously called "Nipissing" dykes, but are relabelled herein as Biscotasing dykes on the basis of paleomagnetism. As with Senneterre dykes, blocking temperatures indicate that the Biscotasing magnetization is carried by low-titanium titanomagnetite (Fig. 5*c*). Although only slightly different in direction from that reported for much younger (1141 Ma) Abitibi dykes (Ernst and Buchan 1993), this remanence is primary in Biscotasing dykes as demonstrated by baked contact tests at four sites from two widely separated localities (see discussion below).

There is a general correlation between dyke trend and direction of remanent magnetization. North-northeast-trending dykes commonly carry the up-north (down-south) Senneterre direction, whereas northeast-trending dykes usually have a down-west Biscotasing direction (Fig. 1). One northeast-trending dyke (sites 16 to 18), however, carries the up-north direction.

An important exception to the pattern of magnetic directions observed in most dykes of this study occurs in sampling sites along north-northeast-trending dykes U1 and U2 of Fig. 2. Individual samples from both dykes are usually stably magnetized to the south-southwest (Fig 6). Stable magnetizations in dyke U1 are streaked from the shallow south-southeast direction towards that of the present Earth's field (Fig. 6*a*), suggesting that a viscous component parallel to the present Earth's magnetic field has not been completely demagnetized in some samples. In a few samples north-northeasterly directions are observed. One sample (see Fig. 6*b*) exhibits a swing between normal and reversed end points. Dyke U1, sampled in this study at four sites and at other sites by Larochelle (1966), can be traced on the basis of its paleomagnetic signature for a distance of 60 km. The remanence of this dyke was used previously to define the magnetic direction of "Preissac" dykes (Goodwin et al. 1972).

The south-southwest magnetization of dykes U1 and U2 is reminiscent of that observed in north- and northwest-trending Matachewan dykes farther to the west (Bates and Halls 1991), which have been dated at 2452^{+3}_{-2} Ma (Heaman 1989). Thus, dykes U1 and U2 may belong to a previously unidentified north-northeast-trending subswarm of the Matachewan dyke swarm. The present data, however, are not sufficient for a definite correlation.

Stable site means for Senneterre and Biscotasing dykes are listed in Tables 2 and 3, respectively. To facilitate comparison

TABLE 2. Paleomagnetic results for 2214 Ma Senneterre dyke swarm

Site	Dyke	Lat. (°N)	Long. (°W)	$N(n)$	D (°)	I (°)	D' (°)	I' (°)	k	α_{95} (°)
14	S1	48.59	78.15	4(3)	195.7	52.9	194.8	53.6	63	15.7
14C[a]	S1	48.59	78.15	9(8)	184.8	45.1	183.9	45.8	60	7.2
16	S2	48.30	78.18	5(5)	353.2	−47.0	352.3	−47.2	177	5.8
17	S2	48.37	78.03	5(5)	359.5	−49.6	358.4	−50.0	756	2.8
18	S2	48.57	77.70	5(5)	356.1	−50.4	354.6	−51.0	111	7.3
23	S3	48.22	78.34	5(5)	348.3	−45.1	347.6	−45.3	300	4.4
24	S4	48.22	78.34	5(4)	358.5	−40.2	357.8	−40.4	195	6.6
25	S5	48.22	78.33	5(5)	353.9	−45.9	353.2	−46.1	516	3.4
29	S6	48.05	77.45	5(5)	359.7	−31.1	358.2	−30.9	34	13.4
30	S6	48.19	77.36	5(5)	359.1	−42.3	357.4	−42.4	18	18.4
31	S6	48.42	77.22	5(5)	1.0	−44.5	359.1	−45.0	150	6.3
32	S6	48.52	77.17	5(4)	3.6	−39.1	1.7	−39.9	124	8.3
33	S6	48.82	76.99	5(4)	350.4	−33.5	348.4	−34.2	116	8.6
Dyke S1 mean				2	—	—	188.9	49.8	—	—
Dyke S2 mean				3	—	—	355.0	−49.4	833	4.3
Dyke S6 mean				5	—	—	356.8	−38.6	131	6.7
Mean of 6 dykes[b,c]				6	—	—	356.4	−45.1	147	5.5
Mean of 13 sites[b,d]				13	—	—	357.2	−44.2	101	4.3

NOTES: N, number of samples measured; n, number of samples used in the statistics; D and I, declination and inclination at the sampling site; D' and I', declination and inclination calculated for a common reference locality at 48°N, 79°W; k, precision parameter; α_{95}, radius of the circle of 95% confidence about the mean direction.

[a]Country rocks adjacent to the dyke site of the same number.

[b]Directions from site S1 have been inverted to calculate the mean direction.

[c]Corresponding paleopole for 6 dykes: lat. = 15.3°S, long. = 75.7°W, dm = 7.0°, dp = 4.4°. dm and dp are the semiaxes of the oval of 95% confidence about the mean paleopole.

[d]Corresponding paleopole for 13 dykes: lat. = 16.0°S, long. = 76.4°W, dm = 5.2°, dp = 3.3°.

Table 3. Paleomagnetic results for 2167 Ma Biscotasing dyke swarm

Site	Dyke	Lat. (°N)	Long. (°W)	$N(n)$	D (°)	I (°)	D' (°)	I' (°)	k	α_{95} (°)
01	B1	48.20	81.32	5(5)	278.1	53.6	279.8	52.0	220	5.2
02	B1	48.36	81.07	5(5)	288.2	66.3	289.6	65.2	59	10.1
03	B1	49.04	79.36	5(4)	295.6	69.8	297.0	69.3	58	12.2
04	B2	47.52	81.76	5(4)	266.3	69.0	267.6	67.6	22	20.1
05	B2	48.02	81.26	5(3)	261.9	52.2	263.8	50.6	971	4.0
06	B2	48.47	80.55	5(4)	266.1	65.4	267.8	64.6	39	14.9
06C[a]	B2	48.47	80.55	5(5)	255.1	55.1	256.7	54.2	204	5.4
07	B2	48.61	80.21	5(4)	253.7	54.3	255.2	53.6	101	9.2
08	B2	48.89	79.39	5(3)	248.1	61.0	249.2	61.1	677	4.7
19	B3	48.22	78.37	5(5)	264.5	58.4	264.2	58.8	173	5.8
20	B3	48.27	78.21	5(4)	252.9	61.9	252.5	62.4	44	14.1
21	B3	48.38	77.97	5(4)	264.8	61.9	264.4	62.5	134	8.0
22	B3	48.55	77.67	5(5)	271.5	50.1	270.9	51.0	523	3.4
26	B4	48.12	77.94	5(5)	266.7	57.3	266.0	58.0	164	6.0
36	B5	46.62	82.80	5(5)	243.7	57.6	246.4	54.8	125	6.9
36C[a]	B5	46.62	82.80	3(3)	238.1	56.0	241.0	53.2	37	20.7
37	B6	46.62	82.80	5(5)	291.3	73.7	290.5	72.5	612	3.1
37C[a]	B6	46.62	82.80	4(4)	292.1	59.4	293.1	57.7	19	21.6
Dyke B1 mean				3	—	—	287.1	62.4	67	15.2
Dyke B2 mean				6	—	—	259.6	58.8	109	6.4
Dyke B3 mean				4	—	—	263.7	58.9	146	7.6
Dyke B5 mean				2	—	—	243.7	54.0	—	—
Dyke B6 mean				2	—	—	292.1	65.1	—	—
Mean of 6 dykes[b]				6	—	—	267.0	60.5	70	8.1
Mean of 18 sites[c]				18	—	—	265.9	60.3	60	4.5

NOTES: Same as notes to Table 2.

[a]Country rocks adjacent to the dyke site of the same number.

[b]Corresponding paleopole for 6 dykes: lat. = 27.8°N, long. = 136.7°W, dm = 12.3°, dp = 9.4°.

[c]Corresponding paleopole for 18 sites: lat. = 27.0°N, long. = 136.3°W, dm = 6.8°, dp = 5.2°.

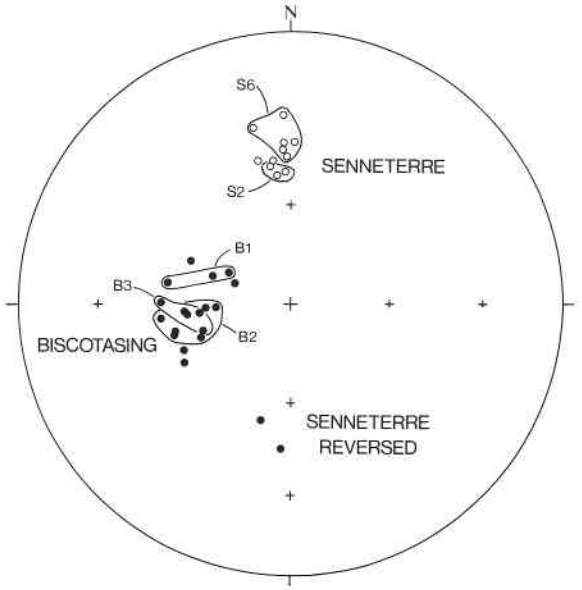


FIG. 7. Mean site directions for Senneterre and Biscotasing dykes and contact host rocks. To facilitate comparison of directions from widely separated geographic localities, directions are converted to a common reference site at 48°N, 79°W. Symbols as in Fig. 4. Clusters of three or more site mean directions from a single dyke are labelled.

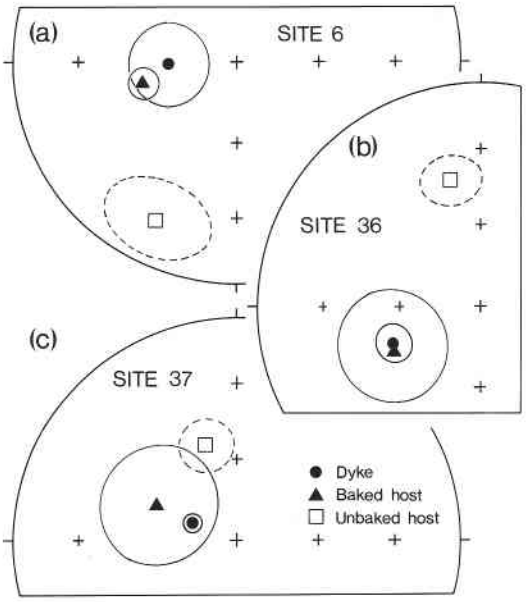


FIG. 8. Positive baked contact tests for Biscotasing dykes at (*a*) site 6, (*b*) site 36, and (*c*) site 37. Data are converted to a reference locality at 48°N, 79°W. Circles of 95% confidence are solid and broken on the lower and upper hemispheres, respectively.

of data from a broad geographic distribution of site localities (Fig. 1), site directions have been recalculated for a common reference locality (48°N, 79°W) and plotted in Fig. 7.

The mean direction for five reversely magnetized and one normally magnetized Senneterre dyke (13 sites) is D = 356.4°, I = −45.1° (precision parameter k = 147, radius of the 95% confidence circle about the mean direction α_{95} = 5.5°), with corresponding paleopole at 15.3°S, 75.7°W (dm = 7.0° and dp = 4.4°, where dm and dp are the semiaxes of the oval of 95% confidence). The mean direction for six normally magnetized Biscotasing dykes (18 sites) is D = 267.0°, I = 60.5° (k = 70, α_{95} = 8.1°), with a corresponding paleopole at 27.8°N, 136.7°W (dm = 12.3°, dp = 9.4°). Because of the non-Fisherian distribution of remanence directions of dykes U1 and U2 , no mean direction was calculated.

Paleomagnetic tests of age of remanence

Stable magnetic remanences of both Senneterre and Biscotasing dykes are interpreted to be primary on the basis of paleomagnetic tests. Tests that are useful for establishing the primary or secondary origin of dyke remanences have been summarized by Buchan and Halls (1990).

Baked contact tests

Positive baked contact tests, which establish that the remanence of intrusive rocks is primary (criteria in Buchan and Halls 1990), have previously been reported at site 7 (see Fig. 2) in a Biscotasing dyke (Irving and Naldrett 1977; Schutts and Dunlop 1981; Buchan and Schwarz 1981). Buchan and Schwarz (1981) describe detailed sampling profiles perpendicular to the dyke contact and demonstrate that overprinting in the contact zone is a thermoremanent magnetization, strengthening the case for a primary dyke remanence. Positive baked contact tests have also been conducted at site A1 (see Fig 2) in a normally magnetized Senneterre dyke (Symons et al. 1981) and for a N1 Nipissing sill that carries a remanence similar to that of Senneterre dykes (Buchan 1991).

In this study positive baked contact tests are established at three additional localities in Biscotasing dykes (site 6 of Fig. 2 and sites 36 and 37 of Fig. 3). Baked contact tests were also attempted for Senneterre dykes S1 and S6 carrying down-south and up-north magnetization directions, respectively, but proved inconclusive because volcanic country rock far from the dyke is unstably magnetized.

At site 6 (Fig. 2), a 90 m thick Biscotasing dyke, B2, cuts a Matachewan dyke. Within 12 m of the Biscotasing dyke contact the Matachewan dyke has acquired a Biscotasing magnetization direction, but samples collected beyond 200 m carry a typical shallow south-southwest Matachewan direction (Fig. 8*a*).

At sites 36 and 37 north of Elliot Lake (Fig. 3) Biscotasing dykes B5 and B6 of thicknesses 2.7 and ~10 m, respectively, cut Nipissing diabase sills. In each case samples were collected along profiles perpendicular to the dyke contacts. Close to the contacts the host Nipissing diabase carries a Biscotasing overprint direction, but at a distance of several dyke thicknesses from the contact a typical up-north Nipissing N1 magnetization is observed (Figs. 8*b* and 8*c*). The Nipissing remanence direction at these sites is indistinguishable from that observed in Senneterre dykes.

Each of the above baked contact tests is positive. Thus, the remanence of Biscotasing dykes is primary at two widely separated localities. Similarily, the earlier baked contact test for a Senneterre dyke (Symons et al. 1981) establishes that the remanence of this swarm is also primary. The positive baked contact tests for Biscotasing dykes which cut Nipissing sills at sites 36 and 37 also demonstrate that the Biscotasing dykes and their magnetization postdate the N1 Nipissing sills and their up-north magnetization.

Secular variation test

Dyke remanences may be shown to be primary if secular variation is detected between dykes that occur in the same geo-

TABLE 4. U–Pb analytical data

Fraction[a]	Weight (mg)	U (ppm)	Pb[b] (ppm)	$^{206}Pb^{c}/^{204}Pb$	$^{208}Pb^{b}$ (%)	$^{206}Pb^{d}/^{238}U$	$^{207}Pb^{d}/^{235}U$	$^{207}Pb^{d}/^{206}Pb$	$^{207}Pb/^{206}Pb$ age[e]
Biscotasing dyke (48°36.6′N; 80°12.6′W)									
A. b,−74,15	0.022	399	150	4 626	4.4	0.36942(0.08)	6.7540(0.10)	0.13260(0.03)	2132.7(1.2)
B. z,+105,a,4	0.013	1423	706	44 160	22.1	0.39505(0.09)	7.3464(0.10)	0.13487(0.03)	2162.3(1.0)
C. z,+105,a,7	0.014	639	305	22 570	20.0	0.39004(0.08)	7.2269(0.10)	0.13438(0.03)	2156.0(1.0)
D. b,+74 −105,1	0.003	679	264	1 407	1.3	0.39244(0.10)	7.2874(0.13)	0.13468(0.06)	2159.9(2.1)
E. b,−74,20	0.005	529	204	885	4.7	0.37674(0.10)	6.9142(0.15)	0.13311(0.10)	2139.4(3.3)
Senneterre dyke (48°11.4′N; 77°21.6′W)									
F. b,−44,35	0.003	208	85	241	2.3	0.40856(0.21)	7.8281(0.46)	0.13896(0.36)	2214.3(12.5)

[a]b, baddeleyite; z, zircon; a, abraded; sizes (e.g., −74 +62) refer to length aspect of zircons in microns (i.e., through 74 μm sieve but not the 62 μm sieve); number refers to number of grains analysed.
[b]Radiogenic Pb.
[c]Measured ratio, corrected for spike and fractionation.
[d]Corrected for blank Pb and U and common Pb (errors quoted are 1σ in percent).
[e]Corrected for blank and common Pb (errors are 2σ in Ma).

graphic locality (e.g., Halls 1986). Sampling sites along a single dyke, which would have cooled rapidly following intrusion, should carry a consistent direction of magnetization, distinct from that carried by sites along another dyke that intruded at a slightly different time.

In the case of Senneterre dykes, sites 16 to 18 along dyke S2 form a cluster of directions (Fig. 7) whose mean is distinct from that for dyke S6 (sites 29 to 33) at the 95% confidence level. This difference undoubtedly records secular variation of the Earth's magnetic field and suggests that the remanence is primary.

Biscotasing dyke B1 (sites 1 to 3) directions have higher declinations than those of dykes B2 (sites 4 to 8) and B3 (sites 19 to 22), as evident in Fig. 7. Mean directions, however, are not distinct at the 95% confidence level, so the case for secular variation is not as convincing as with Senneterre dykes. Nevertheless, secular variation may account for the difference in declination observed between adjacent and parallel dykes B1 and B2 (see Fig. 2), which are separated by about 20 km and were sampled over a distance of 200 km.

Distinct directions in nearby dykes of different ages

A primary age for the Biscotasing remanence is demonstrated by its consistency over a distance of 50 km in dyke B3, whereas an adjacent and subparallel Senneterre dyke S2 carries a distinct direction of magnetization. Geochronologic studies described below indicate that Senneterre dykes are older than Biscotasing dykes. As blocking temperatures are similar in the two dykes, a regional post-Biscotasing overprint would not be expected to affect only one of these dykes along its length.

U–Pb dating studies

In many of the more altered samples that were collected for U–Pb dating, baddeleyite has been partially or wholly pseudomorphed by secondary zircon. Because of the typical fine grain size and delicate grain morphology (very thin needles and blades) of igneous baddeleyite, the amount of baddeleyite recovered from dating samples is commonly very low.

We recovered datable baddeleyite and zircon from one Biscotasing dyke sample at site 7 and one Senneterre dyke sample at site 30. Analytical data are given in Table 4 and plotted on a conventional concordia plot in Fig. 9.

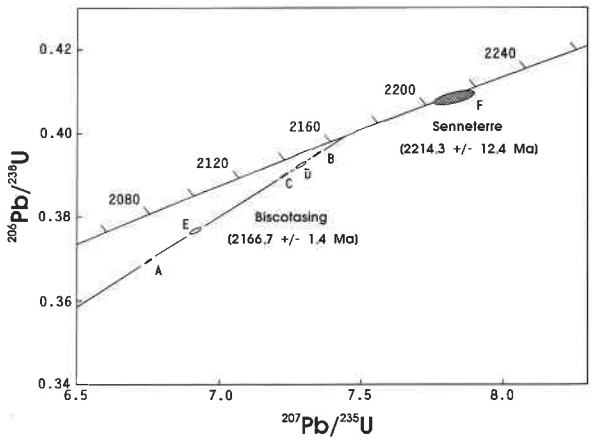


FIG. 9. U–Pb concordia plot for zircon and baddeleyite from Biscotasing and Senneterre dyke samples at sites 7 and 30 respectively. A–F, zircon or baddeleyite fractions of Table 4. Errors are shown at the 2σ level.

Three fractions of baddeleyite and two fractions of zircon were analysed from the Biscotasing dyke sample. The zircons formed clear, colorless fragments of originally simple, square, elongate prisms with clear inclusions and poorly developed cleavage parallel to the *c*-axis. This grain morphology is common for igneous zircon in mafic rocks, and the zircon in the sample is considered to be magmatic, rather than xenocrystic, in origin. The baddeleyite consisted entirely of broken cleavage fragments. One of the baddeleyite fractions was a single, relatively large cleavage fragment weighing approximately 2 μg. The surface of this grain appeared to be slightly frosted, possibly indicating that the grain was partially altered to secondary zircon. The five fractions range from 1.6 to 11.0% discordant and form a linear array (probability of fit = 49% using the Davis (1980) model) with upper and lower intercepts of 2166.7 ± 1.4 Ma and 763 Ma, respectively. Regressing only the two zircon analyses gives essentially the same intercept ages. The baddeleyite fractions are more discordant than is typical for fresh, unaltered baddeleyite (e.g., LeCheminant and Heaman 1989); this may indicate that all of the baddeleyite grains included minor zircon alteration. The lower intercept is

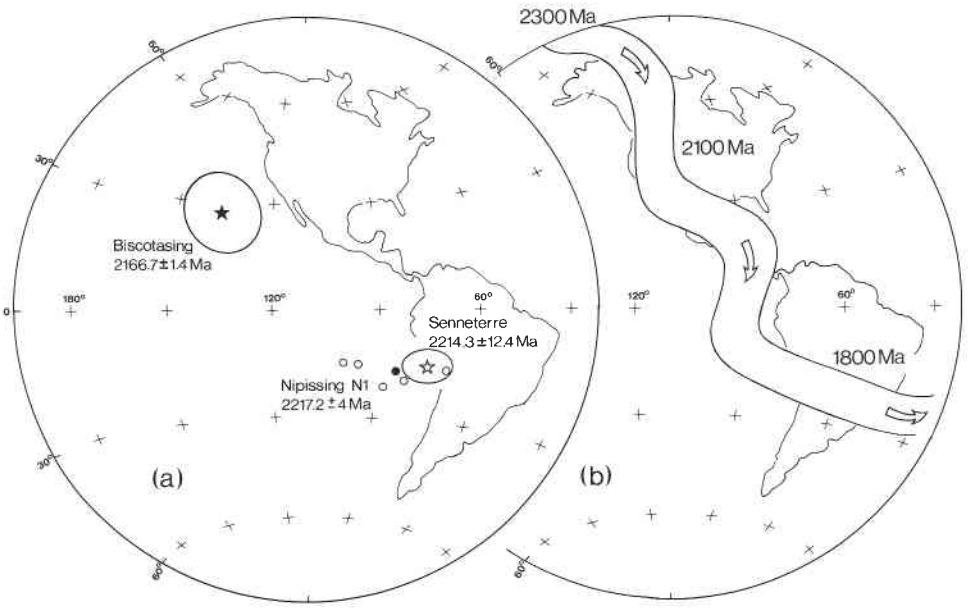


FIG. 10. Comparison of (*a*) well-dated primary paleopoles of Early Proterozoic age for Senneterre dykes, N1 Nipissing sills and Biscotasing dykes of the southern Superior and Southern provinces with (*b*) an example of the most widely used apparent polar wander path for North America for this period (after Irving 1979). In *a* Nipissing N1 poles are shown as circles. They are reported in Symons (1970, 1971), Patel and Palmer (1974), Symons and Londry (1975), Roy and Lapointe (1976), and Stupavsky and Symons (1982). Senneterre and Biscotasing poles of the present study are shown as stars along with their circle of 95% confidence. Closed and open symbols indicate dominantly normal and reversed poles, respectively. U–Pb ages for the Senneterre and Biscotasing dykes are from this study; that for the Nipissing sills is from Noble and Lightfoot (1992).

similar to that obtained by other workers for baddeleyite U–Pb discordia arrays for samples of Nipissing Diabase in the Southern Province (e.g., Corfu and Andrews 1986; Noble and Lightfoot 1992). We therefore consider the upper intercept to give the best estimate of the crystallization age of the dyke. It confirms the $^{40}Ar/^{39}Ar$ age of 2150 ± 50 Ma (2σ) reported by Hanes and York (1979) for mafic (dominantly pyroxene) separates from a sample collected at the same site.

Only a very small amount of extremely fine grained baddeleyite was recovered from one Senneterre dyke sample. The analysis is concordant, however, with a relatively imprecise $^{207}Pb/^{206}Pb$ age of 2214.3 ± 12.4 Ma. We consider this to be the crystallization age of the Senneterre dyke, although further work will be necessary if the age is to be better constrained.

Discussion

Northeast-trending quartz tholeiite dykes of the southern Superior Province which previously were labelled as Preissac dykes can be separated into two, or possibly three, distinct sets. The Senneterrre dyke swarm, emplaced at about 2214 Ma, carries a primary remanent magnetization directed upward to the north. This direction does not differ significantly from that of the dominant N1 magnetization reported for Nipissing sills of the Southern Province (e.g., Symons 1971; Patel and Palmer 1974). Although the interpretation of early baked contact tests for the N1 Nipissing remanence has been disputed in the past (e.g., McGlynn and Irving 1981), a recent baked contact test (Buchan 1991) is positive and demonstrates that the N1 remanence is primary. A sampling site carrying an N1 paleomagnetic direction (K.L. Buchan, unpublished data) has been dated and yields a U–Pb baddeleyite age of 2217.2 ± 4 Ma (Noble and Lightfoot 1992). Thus, Senneterre dykes are of about the same age as N1 Nipissing sills. It should be noted that Noble and Lightfoot (1992) report a similar U/Pb age (2209.6 ± 3.5 Ma) for a probable N3 Nipissing paleomagnetic site. Further studies are being conducted to establish the origin of the up-west N3 component which is observed locally in an area north of Cobalt (Buchan et al. 1989).

The 2166.7 ± 1.4 Ma Biscotasing swarm carries a primary remanence directed steeply downward to the west. A similar remanence direction has been reported at some (N2) Nipissing sill sites (e.g., Roy and Lapointe 1976) and in some "Nipissing" dykes north of Elliot Lake (Symons 1971). It is uncertain if the N2 remanence of Nipissing sills is primary. However, the positive baked contact test for "Nipissing" dyke sites 36 and 37 in the present study indicates that the remanence in these dykes is primary and significantly postdates the age of host N1 Nipissing sills. Therefore, "Nipissing" dykes, at least those with a northeasterly trend, belong to the 2167 Ma Biscotasing swarm.

Biscotasing dykes in the present study are confined to the region southeast of the Kapuskasing Structural Zone (Fig. 1). Further paleomagnetic, petrographic, and geochronologic studies are required to establish the extent of the swarm within and north of the Kapuskasing Structural Zone.

Only two dykes of the present study (U1 and U2) have the shallow southwest remanence direction that was originally used to define "Preissac" dykes (see Goodwin et al. 1972). Both have rather scattered magnetization directions, making it difficult to calculate a mean remanence direction. Nevertheless, these directions appear to be distinct from those of the Senneterre and Biscotasing swarms. The age of north-northeast-trending dykes U1 and U2 has yet to be established, although the remanence directions suggest a possible correlation with the 2452 Ma Matachewan dyke swarm.

Paleopoles for the Senneterre and Biscotasing dyke swarms and the N1 Nipissing sills (Fig. 10*a*) are now among the best-dated Early Proterozoic poles from North America. Each has

been established as primary, each has a well-defined U–Pb age, and in each case the radiometric age has been obtained at a paleomagnetic sampling site. Thus, these paleopoles constitute key tie points for the construction of apparent polar wander paths. Unfortunately, other paleopoles for the 2250–2150 Ma period are poorly constrained in age (e.g., Roy 1983; Buchan and Halls 1990), so there are as yet insufficient data to allow the construction of a reliable polar wander path.

The most widely used North American polar wander path, as drawn by Roy and Lapointe (1976), Irving (1979), Hyodo and Dunlop (1989), and others (e.g., Fig. 10*b*), tracks southeast between 2300 and 1800 Ma, as do paths for the Superior Province alone (Irving et al. 1984; Dunsmore and Symons 1990). However, primary paleopoles for ~2217 Ma Nipissing sills and Senneterre dykes and 2167 Ma Biscotasing dykes establish a trend to the northwest with decreasing age (Fig. 10*a*), a trend that is opposite to that of the widely used path of Fig. 10*b*. The younger end of the track of Fig. 10*b* is also suspect, as it was originally drawn on the assumption that the Nipissing N1 paleopole reflects secondary overprinting at about 1800 Ma.

The North American polar wander path of Fig. 10*b* uses data from across the Canadian Shield, even though there is no clear evidence (paleomagnetic or otherwise) to suggest that the Shield moved as a single unit during the Early Proterozoic. Therefore, until well-dated paleopoles are available from the different structural provinces we consider the paleopoles defined in this study as valid only for the southern Superior and Southern provinces, an area known to have remained largely intact since the intrusion of the 2452 Ma Matachewan dyke swarm.

This study demonstrates the importance of conducting paleomagnetic studies and U–Pb dating at the same sampling sites. When more than a single age of intrusions are present, but have not been distinguished on geological criteria, attempting to integrate U–Pb dating and paleomagnetic results obtained from different sampling sites could lead to erroneous conclusions concerning the age of the intrusions and the age of their paleopoles.

As pointed out by Roy (1983) and Buchan and Halls (1990) most paleomagnetic data presently available for the Precambrian of North America are too poorly constrained in time to allow the construction of polar wander paths, or to distinguish relative motion between continental fragments or structural provinces. Such attempts are likely to be of limited value until ages of remanence are better constrained by precise isotopic dating, and until rigorous paleomagnetic tests of the age of remanence are routinely carried out in paleomagnetic studies.

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Bates, M.P., and Halls, H.C. 1991. Broad-scale Proterozoic deformation of the central Superior Province revealed by paleomagnetism of the 2.45 Ga Matachewan dyke swarm. Canadian Journal of Earth Sciences, **28**: 1780–1796.

Buchan, K.L. 1991. Baked contact test demonstrates primary nature of dominant (N1) magnetisation of Nipissing intrusions in Southern Province, Canadian Shield. Earth and Planetary Science Letters, **105**: 492–499.

Buchan, K.L., and Halls, H.C. 1990. Paleomagnetism of Proterozoic mafic dyke swarms of the Canadian Shield. *In* Mafic dykes and emplacement mechanisms. *Edited by* A.J. Parker, P.C. Rickwood, and D.H. Tucker. Balkema, Rotterdam, pp. 209–230.

Buchan, K.L., and Schwarz, E. 1981. Uplift estimated from remanent magnetization: Munro area of Superior Province since 2150 Ma ago. Canadian Journal of Earth Sciences, **18**: 1164–1173.

Buchan, K.L., Card, K.D., and Chandler, F.W. 1989. Multiple ages of Nipissing Diabase intrusion: paleomagnetic evidence from the Englehart area, Ontario. Canadian Journal of Earth Sciences, **26**: 427–445.

Card, K.D. 1981. Progress report on regional geological synthesis, central Superior Province. Part 1. Reconnaissance investigations and studies of igneous intrusions. *In* Current research, part A. Geological Survey of Canada, Paper 81-1A, pp. 78–82.

Card, K.D. 1990. A review of the Superior Province of the Canadian Shield, a product of Archean accretion. Precambrian Research, **48**: 99–156.

Condie, K.C., Bobrow, D.J., and Card, K.D. 1987. Geochemistry of Precambrian mafic dykes from the southern Superior Province of the Canadian Shield. *In* Mafic dyke swarms. *Edited by* H.C. Halls and W.F. Fahrig. Geological Association of Canada, Special Paper 34, pp. 95–108.

Corfu, F., and Andrews, A.J. 1986. A U–Pb age for mineralized Nipissing diabase, Gowganda, Ontario. Canadian Journal of Earth Sciences, **23**: 107–109.

Davis, D.W. 1980. Optimum linear regression and error estimation applied to U–Pb data. Canadian Journal of Earth Sciences, **19**: 2141–2149.

Dunsmore, D.J., and Symons, D.T.A. 1990. Paleomagnetism of the Lynn Lake gabbros in the Trans-Hudson Orogen and closure of the Slave and Superior cratons. *In* The Early Proterzoic Trans-Hudson Orogen of North America. *Edited by* J.F. Lewry and M.R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 215–228.

Ernst, R.E., and Buchan, K.L. 1993. Paleomagnetism of the Abitibi dyke swarm, southern Superior Province and implications for the Logan Loop. Canadian Journal of Earth Sciences, **30**, in press.

Fahrig, W.F. 1987. The tectonic settings of continental mafic dyke swarms: failed arm and early passive margin. *In* Mafic dyke swarms. *Edited by* H.C. Halls and W.F. Fahrig. Geological Association of Canada, Special Paper 34, pp. 331–348.

Fahrig, W.F., Gauther, E.H., and Larochelle, A. 1965. Palaeomagnetism of diabase dykes of the Canadian Shield. Canadian Journal of Earth Sciences, **2**: 278–298.

Gates, T.M., and Hurley, P.M. 1973. Evaluation of Rb–Sr dating methods applied to the Matachewan, Abitibi, Mackenzie, and Sudbury dike swarms in Canada. Canadian Journal of Earth Sciences, **10**: 900–919.

Goodwin, A.M., Ambrose, J.W., Ayers, L.D., Clifford, P.M., Currie, K.L., Ermanovics, I.M., Fahrig, W.F., Gibbs, R.A., Hall, D.A., Innes, M.J.S., Irvine, T.N., MacLaren, A.S., Norris, A.W., and Pettijohn, F.J. 1972. The Superior Province. *In* Variations in tectonic styles in Canada. *Edited by* R.A. Price and R.J.W. Douglas. Geological Association of Canada, Special Paper 11, pp. 527–632.

Halls, H.C. 1986. Paleomagnetism, structure, and longitudinal correlation of Middle Precambrian dykes from northwestern Ontario and Minnesota. Canadian Journal of Earth Sciences, **23**: 142–157.

Hanes, J.A., and York, D. 1979. A detailed $^{40}Ar/^{39}Ar$ age study of an Abitibi dike from the Canadian Superior Province. Canadian Journal of Earth Sciences, **16**: 1060–1070.

Heaman, L.M. 1989. U–Pb dating of mafic dyke swarms: what are the options? *In* International Association of Volcanology and Chemistry of the Earth's Interior General Assembly, Program of

Abstracts. New Mexico Bureau of Mines and Mineral Resources, Bulletin 131, p. 125.

Hyodo, H., and Dunlop, D.J. 1989. Multicomponent magnetizations in Nipissing diabase near Temagami, Ontario, and structural and thermal history of the Grenville Front. Canadian Journal of Earth Sciences, **26**: 467–478.

Irving, E. 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terrains. Canadian Journal of Earth Sciences, **16**: 669–694.

Irving, E., and Naldrett, A.J. 1977. Paleomagnetism in Abitibi greenstone belt, and Abitibi and Matachewan dikes: Evidence of the Archean geomagnetic field. Journal of Geology, **85**: 157–167.

Irving, E., Davidson, A., and McGlynn, J.C. 1984. Paleomagnetism of gabbros of the Early Proterozoic Blachford Lake intrusive suite and the Easter Island dyke, Great Slave Lake, NWT: possible evidence for earliest continental drift. Geophysical Surveys, **7**: 1–25.

Kirschvink, J.K. 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

Krogh, T.E., Corfu, F., Davis, D.W., Dunning, G.R., Heaman, L.M., Kamo, S.L., Machado, N., Greenough, J.D., and Nakamura, E. 1987. Precise U–Pb isotopic ages of diabase dykes and mafic to ultramafic rocks using trace amounts of baddeleyite and zircon. *In* Mafic dyke swarms. Edited by H.C. Halls and W.F. Fahrig. Geological Association of Canada, Special Paper 34, pp. 147–152.

Larochelle, A. 1966. Palaeomagnetism of the Abitibi dyke swarm. Canadian Journal of Earth Sciences, **3**: 671–683.

LeCheminant, A.N., and Heaman, L.M. 1989. Mackenzie igneous events, Canada: Middle Proterozoic hotspot magmatism associated with ocean opening. Earth and Planetary Science Letters, **96**: 38–48.

McGlynn, J.C., and Irving, E. 1981. Horizontal motions and rotations in the Canadian Shield during the Early Proterozoic. *In* Proterozoic basins of Canada. *Edited by* F.H.A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 183–190.

Ministère de l'Énergie et des Ressources du Québec. 1981*a*. Gites mineraux du Quebec, Region de l'Abitibi, Feuille Lac Waswanipi 32 F. Ministère de l'Énergie et des Ressources du Québec, Map M-306, scale 1 : 250 000.

Ministère de l'Énergie et des Ressources du Québec. 1981*b*. Gites mineraux du Quebec, Region de l'Abitibi, Feuille Rouyn–Noranda 32 D. Ministère de l'Énergie et des Ressources du Québec, Map M-308, scale 1 : 250 000.

Ministère de l'Énergie et des Ressources du Québec. 1981*c*. Gites mineraux du Quebec, Region de l'Abitibi, Feuille Senneterre 32 C. Ministère de l'Énergie et des Ressources du Quebec, Map M-309, scale 1 : 250 000.

Mortensen, J.K. 1993. U–Pb geochronology of the eastern Abitibi Subprovince. Part 1: Chibougamau–Matagami–Joutel region. Canadian Journal of Earth Sciences, **30**: 11–28.

Noble, S.R., and Lightfoot, P.C. 1992. U–Pb baddeleyite ages of the Kerns and Triangle Mountain intrusions, Nipissing Diabase, Ontario. Canadian Journal of Earth Sciences, **29**: 1424–1429.

Ontario Geological Survey. 1991. Bedrock geology of Ontario, east-central sheet. Ontario Geological Survey, Map 2543, scale 1 : 1 000 000.

Patel, J.P., and Palmer, H.C. 1974. Magnetic and paleomagnetic studies of the Nipissing diabase, Lake Matinenda area, Ontario. Canadian Journal of Earth Sciences, **11**: 353–361.

Roy, J.L. 1983. Paleomagnetism of the North American Precambrian: A look at the data base. Precambrian Research, **19**: 319–348.

Roy, J.L., and Lapointe, P.L. 1976. The paleomagnetism of Huronian red beds and Nipissing diabase; post-Huronian igneous events and apparent polar path for the interval −2300 to −1500 Ma for Laurentia. Canadian Journal of Earth Sciences, **13**: 749–773.

Schutts, L.D., and Dunlop, D.J. 1981. Proterozoic magnetic overprinting of Archean rocks in the Canadian Superior Province. Nature (London), **291**: 642–645.

Strangway, D.W. 1961. Magnetic properties of diabase dykes. Journal of Geophysical Research, **66**: 3021–3032.

Stupavsky, M., and Symons, D.T.A. 1982. Extent of Grenvillian remanence components in rocks of the Southern Province. Canadian Journal of Earth Sciences, **19**: 698–708.

Symons, D.T.A. 1970. Paleomagnetism of the Nipissing diabase, Cobalt area, Ontario. Canadian Journal of Earth Sciences, **7**: 86–90.

Symons, D.T.A. 1971. A paleomagnetic study of the Nipissing diabase, Blind River – Elliot Lake area, Ontario. Geological Survey of Canada, Paper 70-63, pp. 19–30.

Symons, D.T.A., and Londry, J.W. 1975. Tectonic results from paleomagnetism of the Aphebian Nipissing Diabase at Gowganda, Ontario. Canadian Journal of Earth Sciences, **12**: 940–948.

Symons, D.T.A., Quick, A.W., and Stupavsky, M. 1981. Magnetic and paleomagnetic characteristics of the Archean iron formation and host rocks at the Adams Mine, Ontario. Ontario Geological Survey, Miscellaneous Paper 98, pp. 298–307.